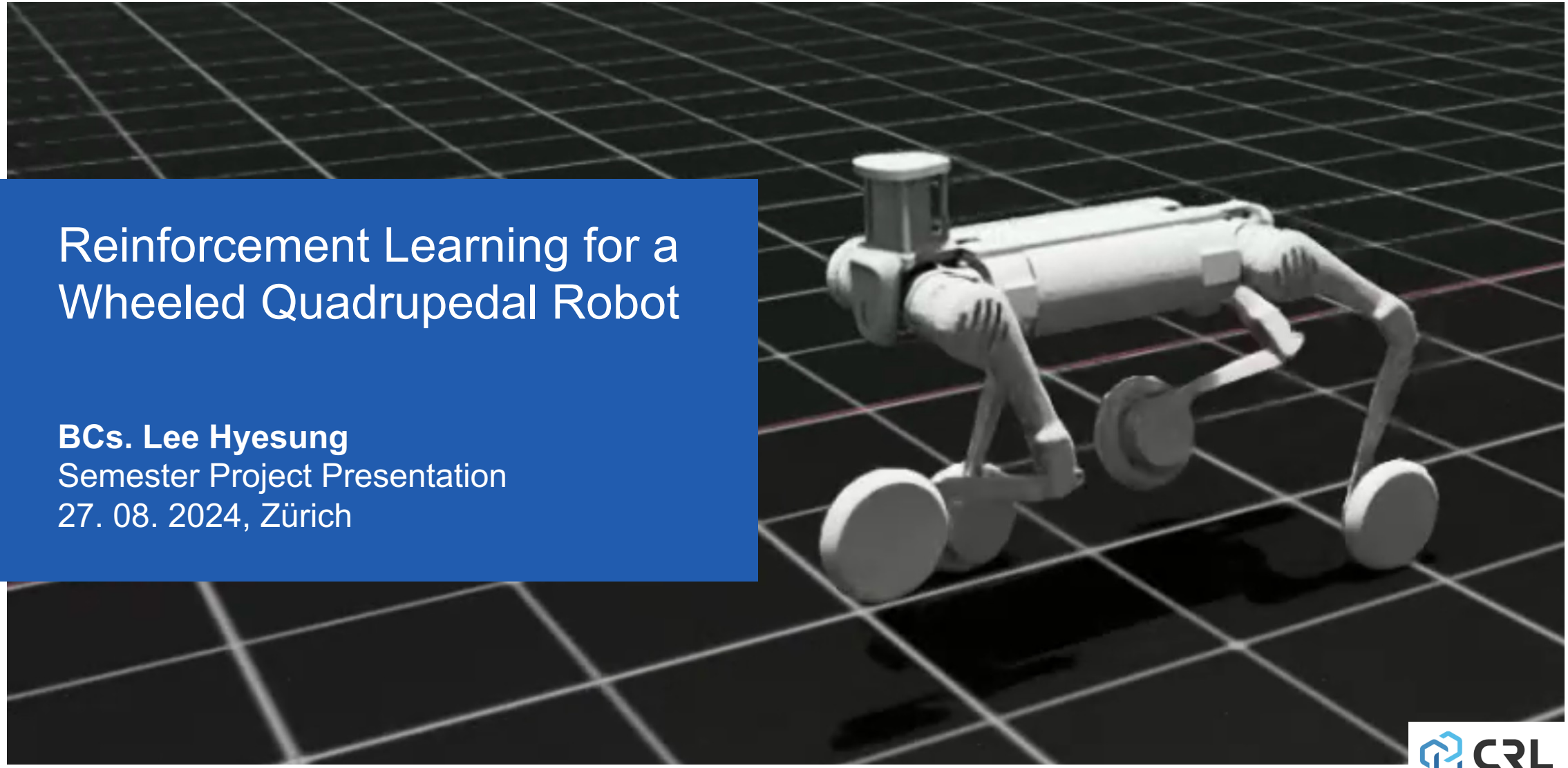
ETH zürich

Computational Robotics Lab



Reinforcement Learning for a Wheeled Quadrupedal Robot

BCs. Lee Hyesung

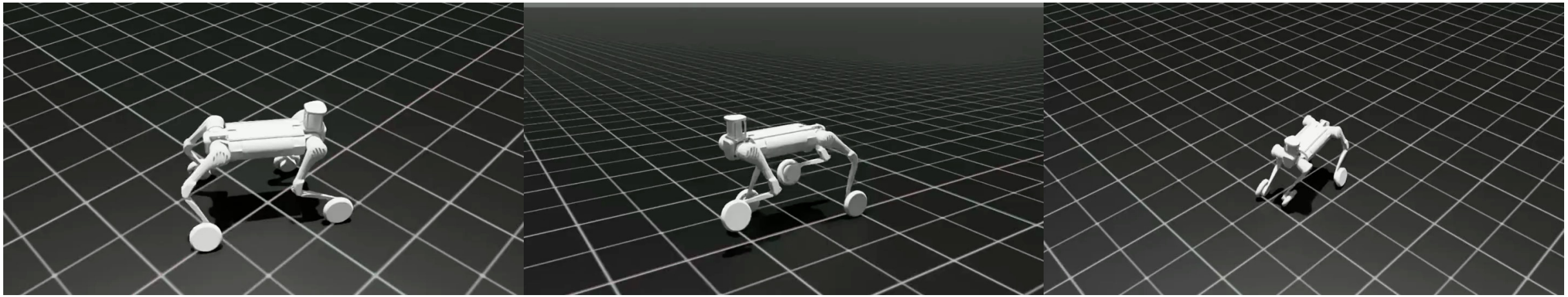
Semester Project Presentation

27. 08. 2024, Zürich

CRL Computational Robotics Lab

Train Wheel-Legged Robot in Simulation

Robot Behaviors



Motivation

Wheel - Legged Locomotion by RSL





1. Learning Robust Autonomous Navigation and Locomotion for Wheeled-legged Robots (RSL)
   https://www.youtube.com/watch?v=7uwxqMm5bnc

2. This four-legged robot on wheels can walk like a human. (RSL)
   https://www.youtube.com/watch?v=vJXQG2_85V0

Motivation

Advantage of wheel legged locomotion

1. Able to overcome **rough terrain**
2. Moving **faster** than the legged locomotion
3. Energy **efficient**



1. Wheels Are Better Than Feet for Legged Robots (IEEE Spectrum)
   https://spectrum.ieee.org/wheels-are-better-than-feet-for-legged-robots

Why Reinforcement Learning?

The reasons of using reinforced learning

1. **Complicate to model** the wheeled legged locomotion

2. **Generate** various motions by trials

3. **Robust** locomotion

Principal Reference Papers

Lee et al. 2020

SCIENCE ROBOTICS | RESEARCH ARTICLE

ANIMAL ROBOTS

Learning quadrupedal locomotion over challenging terrain

Joonho Lee[1]*, Jemin Hwangbo[1,2], Lorenz Wellhausen[1], Vladlen Koltun[3], Marco Hutter[1]

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Miki et al. 2022

SCIENCE ROBOTICS | RESEARCH ARTICLE

ANIMAL ROBOTS

Learning robust perceptive locomotion for quadrupedal robots in the wild

Takahiro Miki[1]*, Joonho Lee[1], Jemin Hwangbo[2], Lorenz Wellhausen[1], Vladlen Koltun[3], Marco Hutter[1]

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Lee et al. 2024

SCIENCE ROBOTICS | RESEARCH ARTICLE

Check for updates

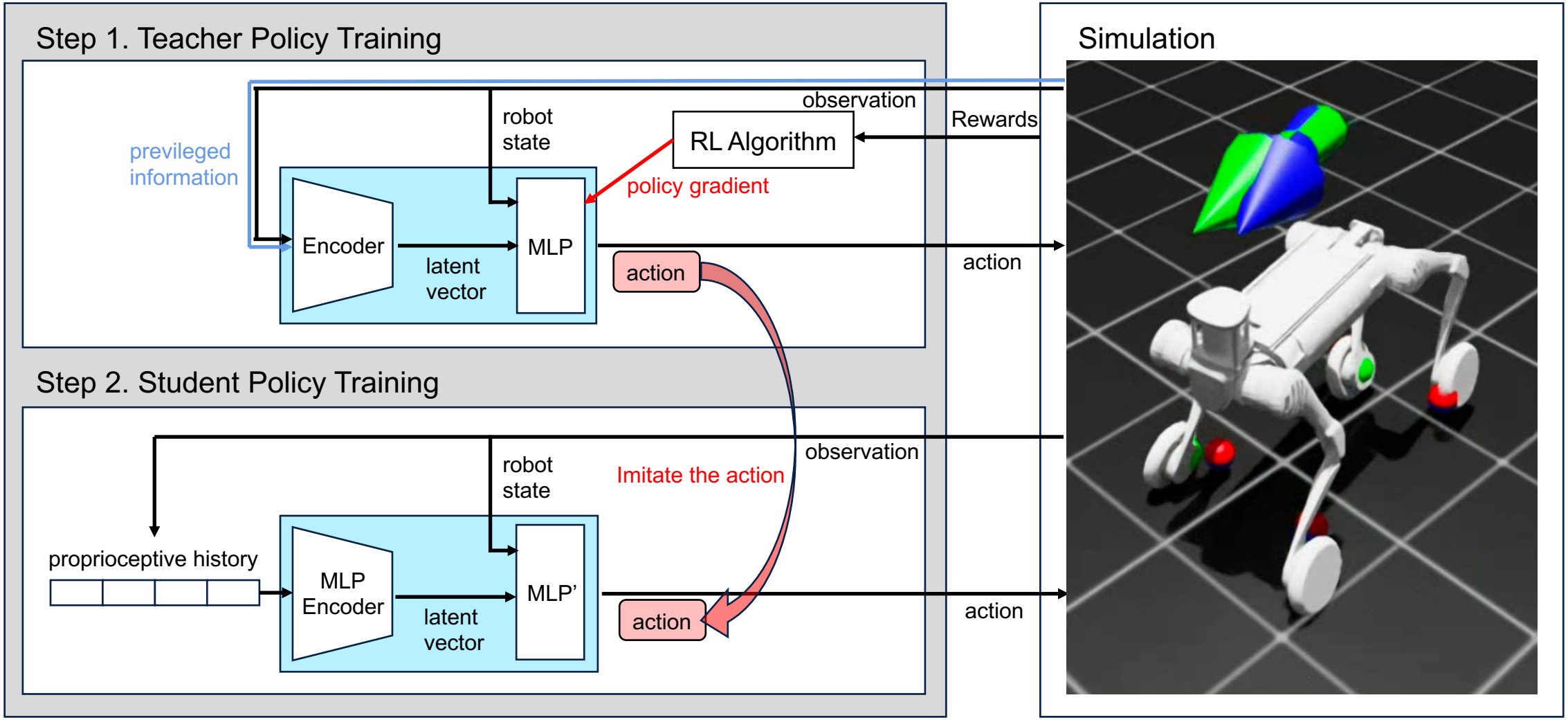
AUTONOMOUS VEHICLES

Learning robust autonomous navigation and locomotion for wheeled-legged robots

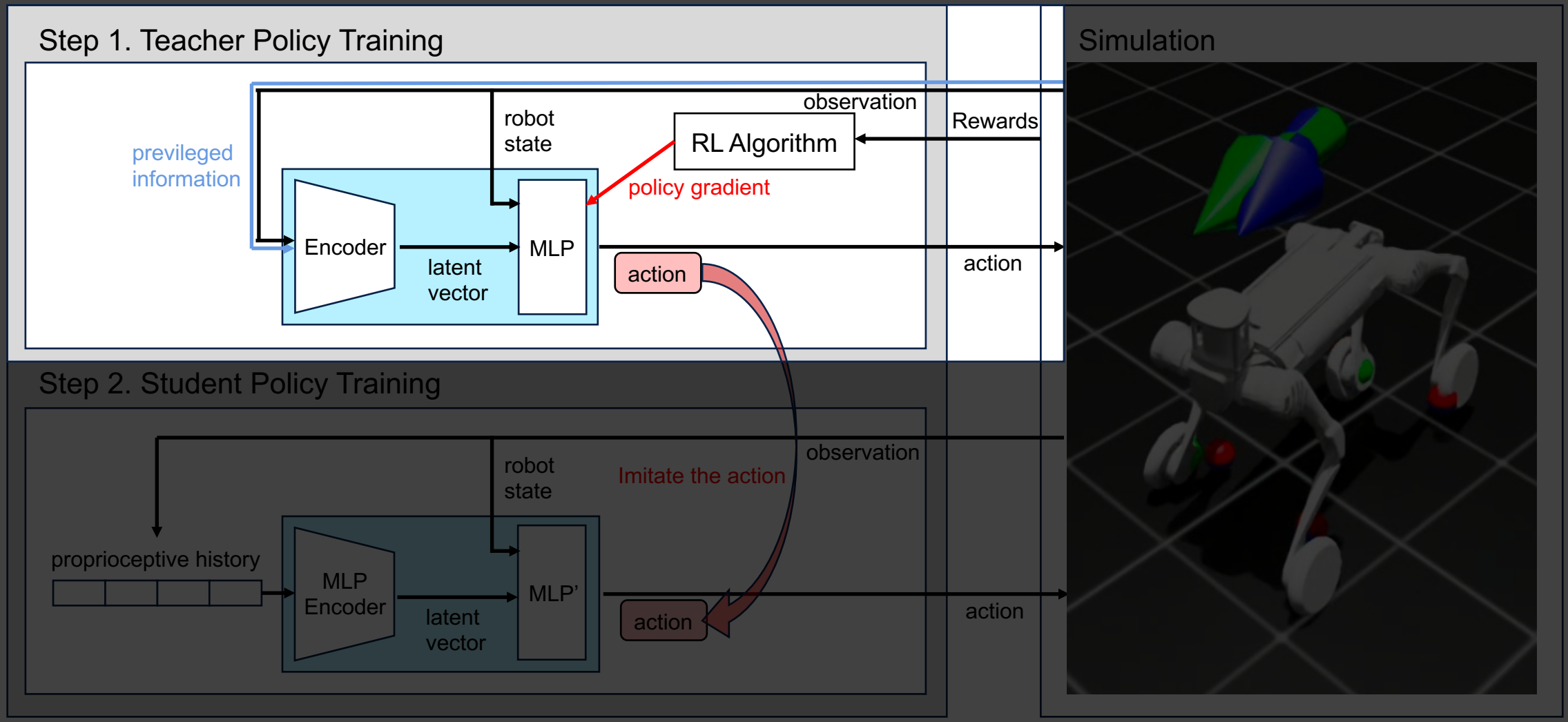
Joonho Lee*†, Marko Bjelonic‡, Alexander Reske‡, Lorenz Wellhausen‡, Takahiro Miki, Marco Hutter

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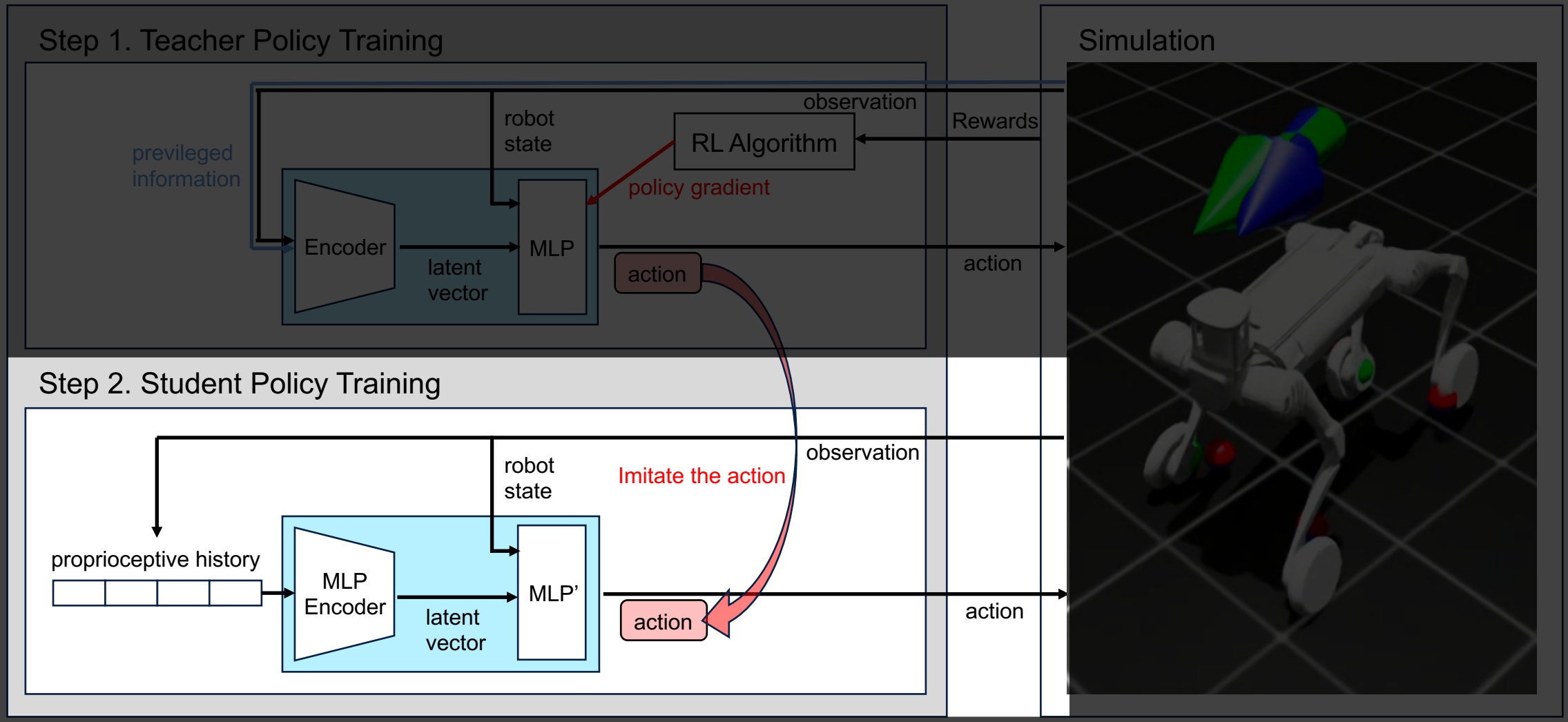
Overall Structure of Previous Works



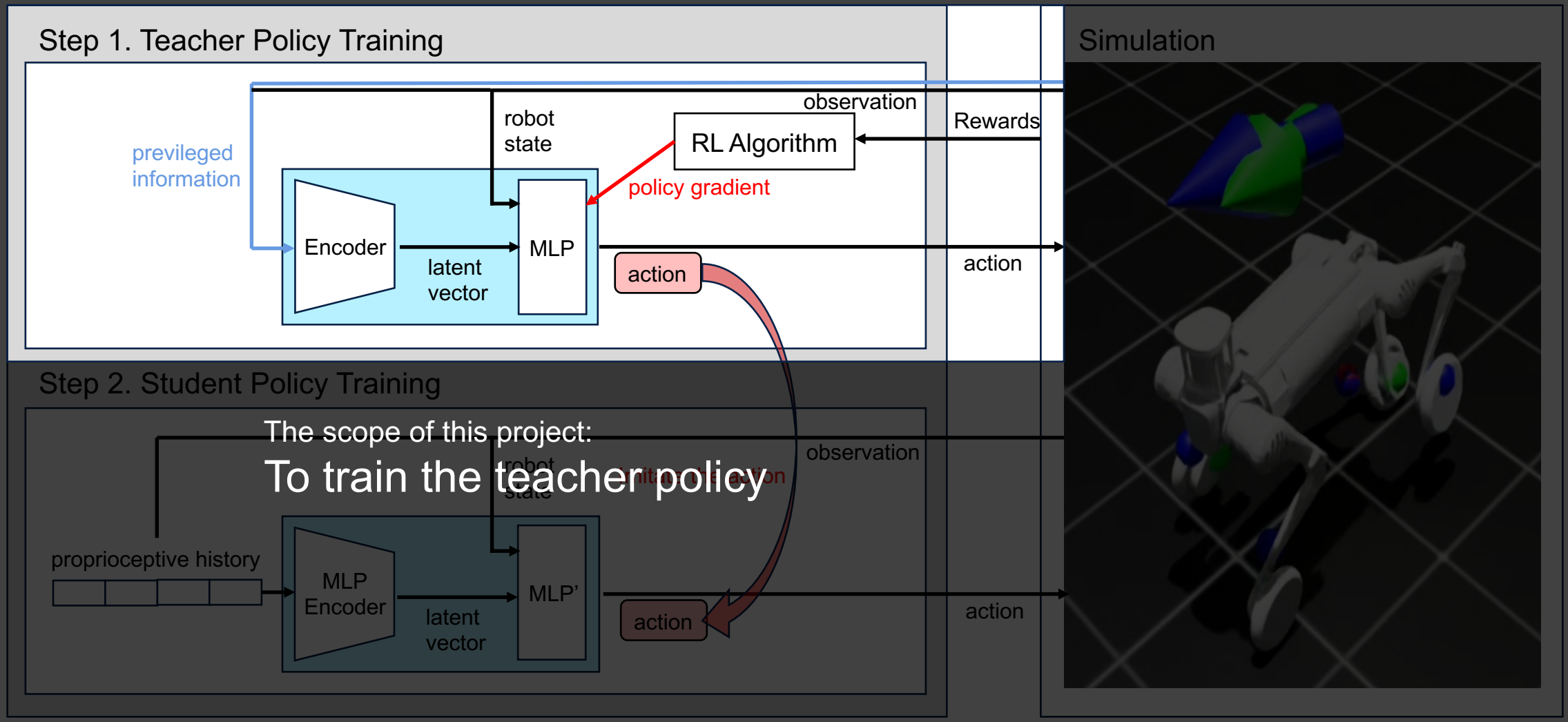
Overall Structure of Previous Works



Overall Structure of Previous Works



Overall Structure of Previous Works



Action Space

The structure how the policy controls the robot

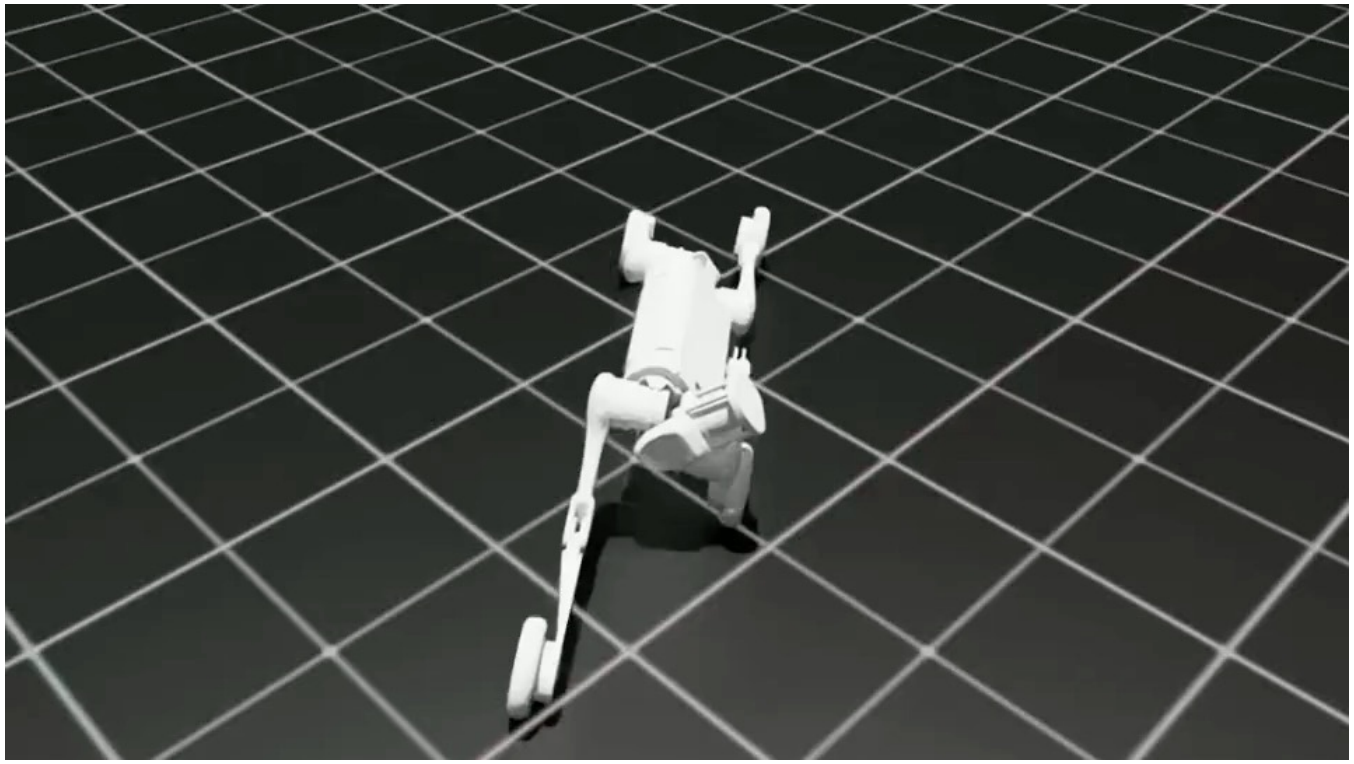
Results: Trained Robot Behavior
Direct Joint Control



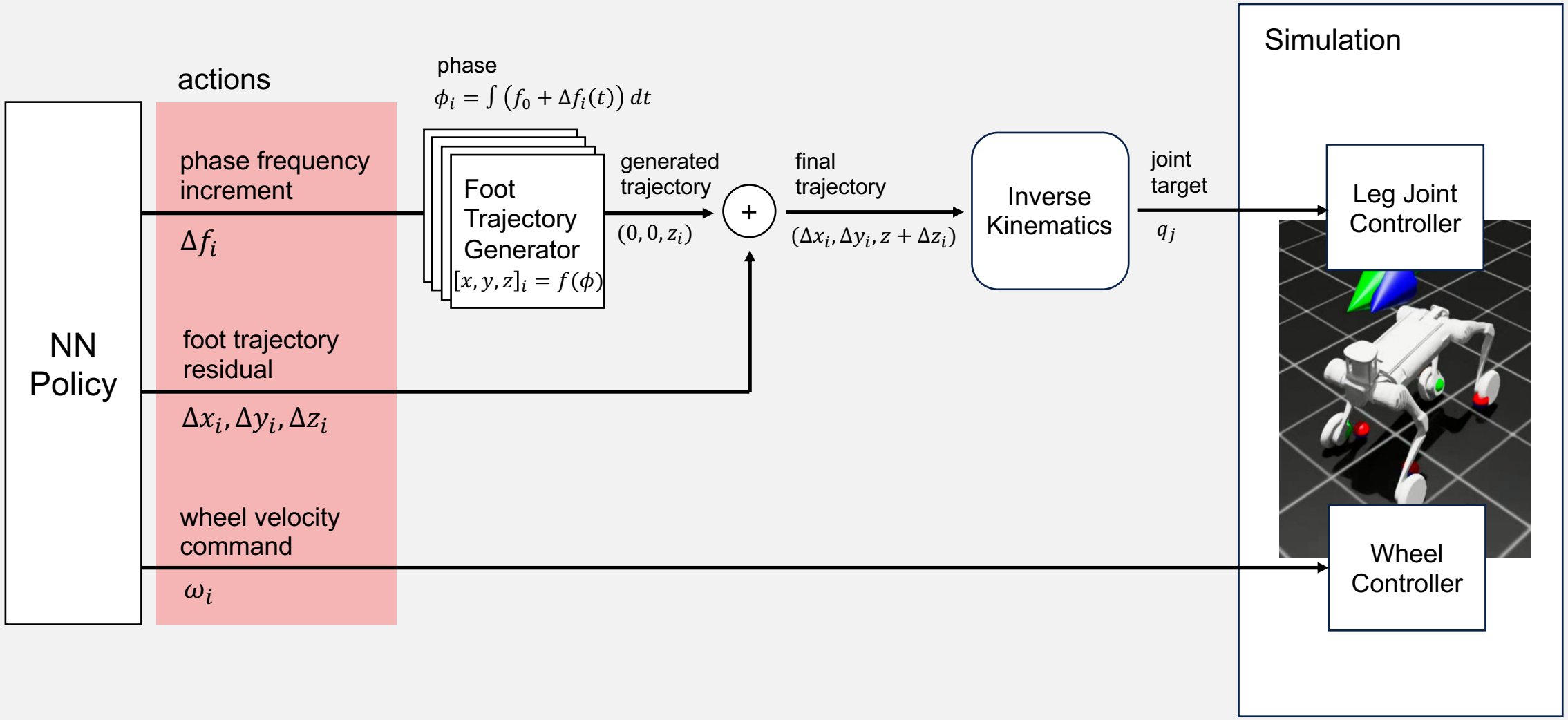
1. Learning robust autonomous navigation and locomotion for wheeled-legged robots, Lee et al. (2024)

Results: Trained Robot Behavior
Direct Joint Control

Direct Joint Control

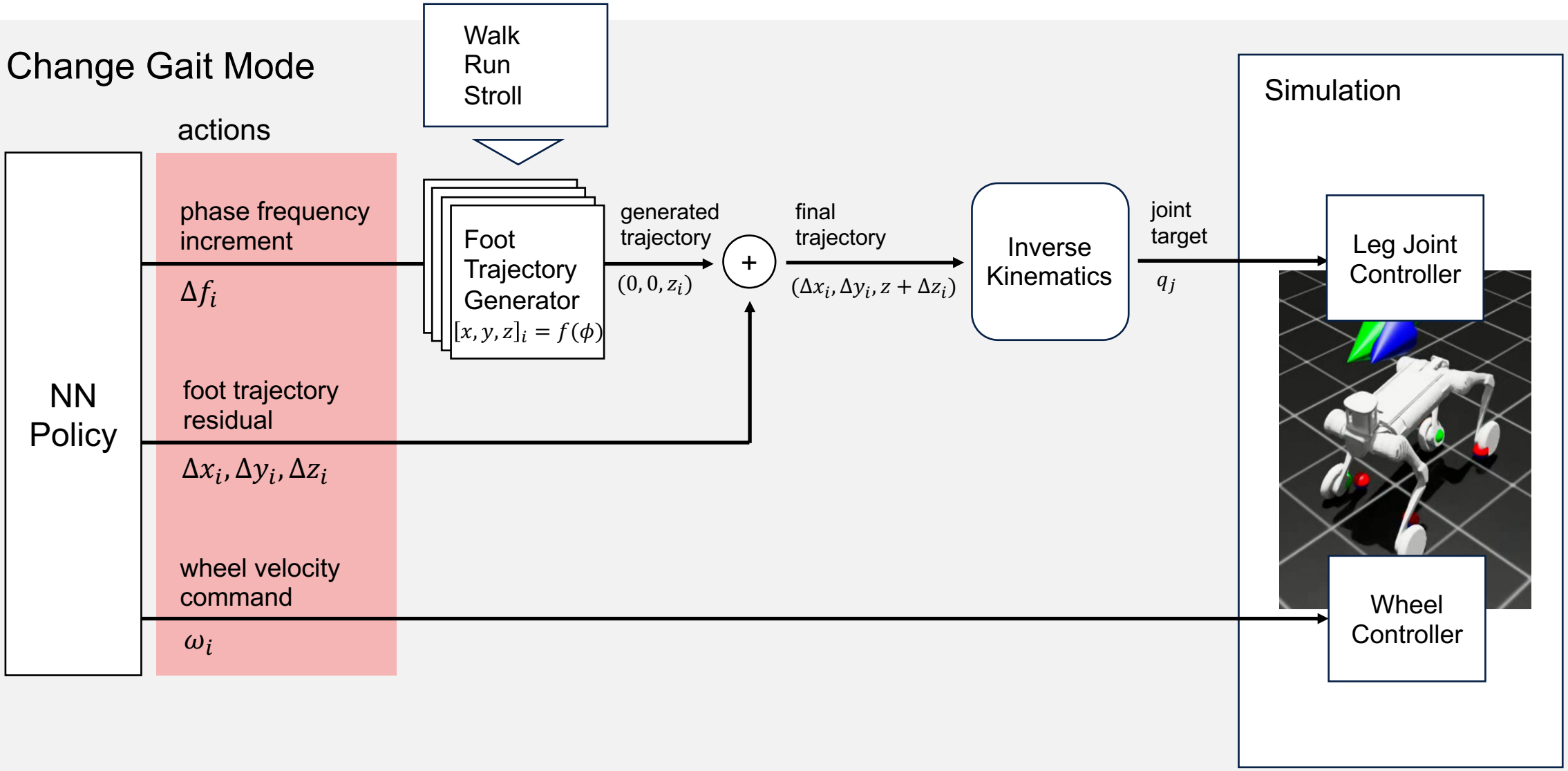


Control Structure



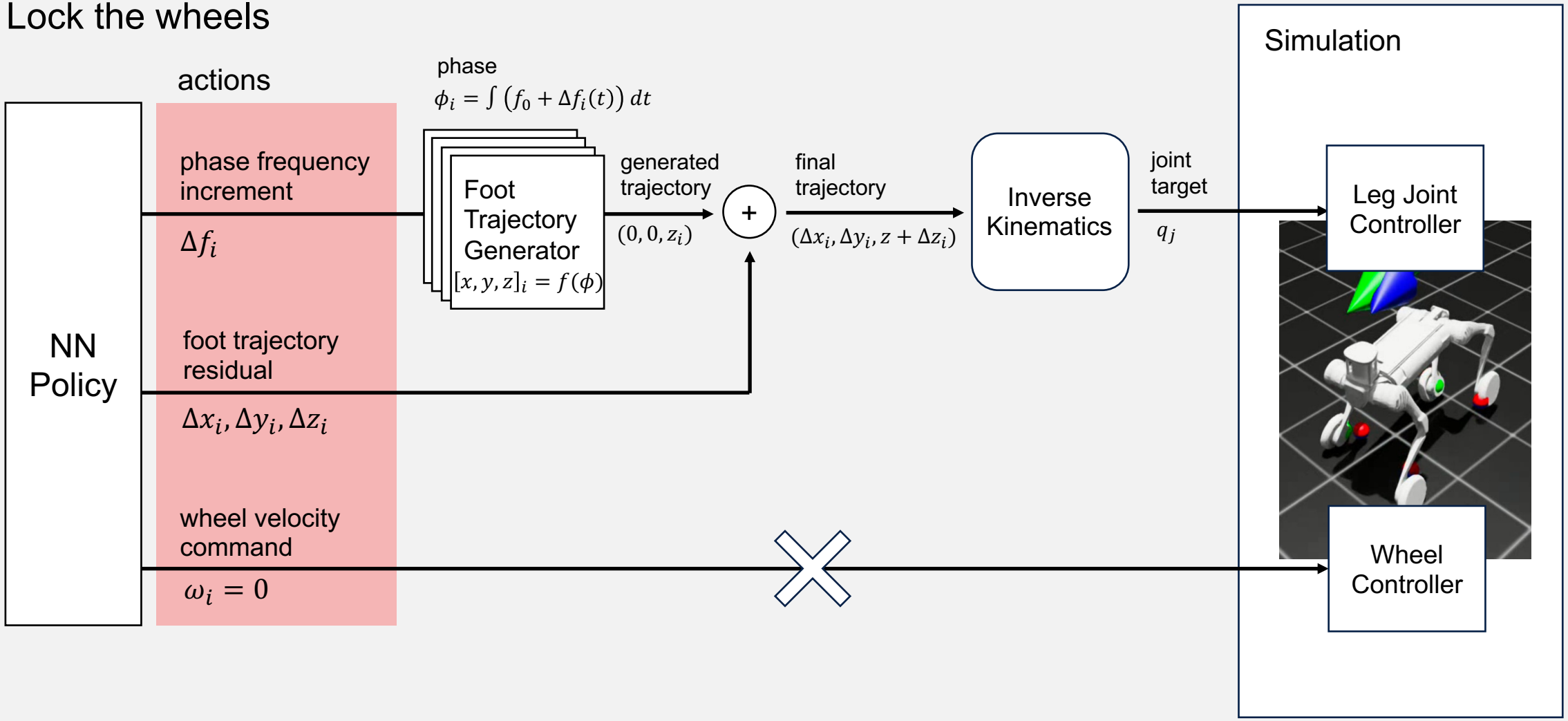
1. Learning quadrupedal locomotion over challenging terrain, Lee et al. (2020)
2. Learning robust perceptive locomotion for quadrupedal robots in the wild, Miki et al. (2022)

Control Structure

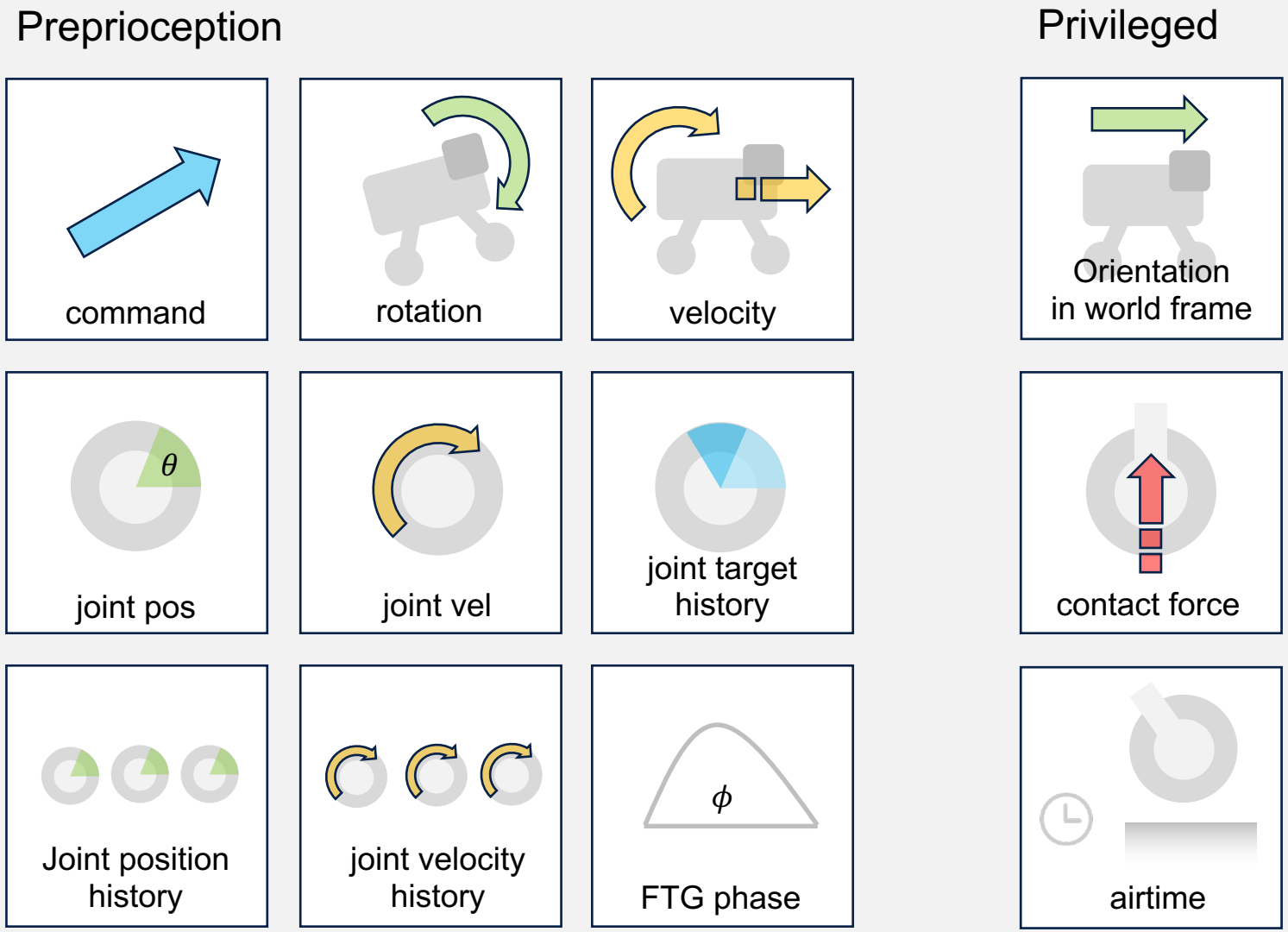


Control Structure

Lock the wheels

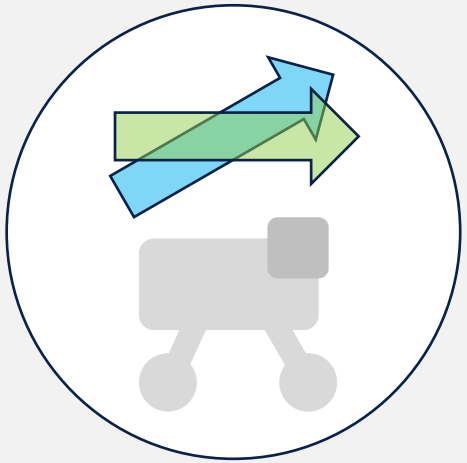


Observation space



Rewards

Command



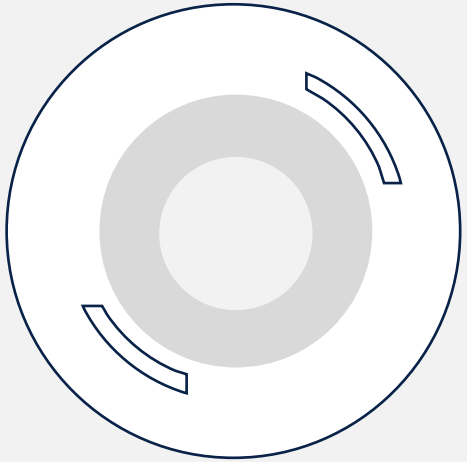
- Linear Velocity Reward
- Yaw Velocity Reward

Stability



- Base Motion Penalty
- Base Orientation Reward
- Base Height Reward
- (Airtime Penalty)

Regularization



- Torque Penalty
- Joint Vibration Penalty
- Joint Target Smoothness Penalty

Termination

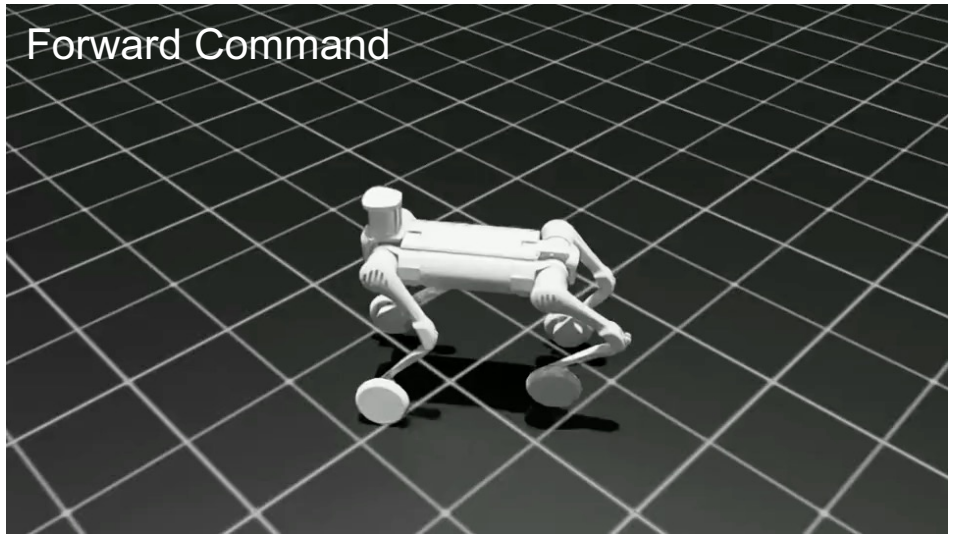


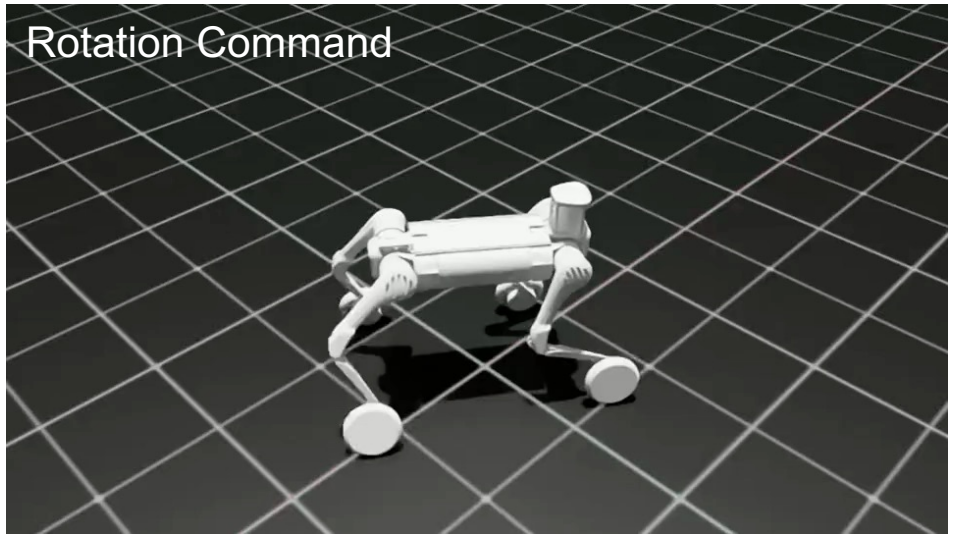
- No-termination Reward

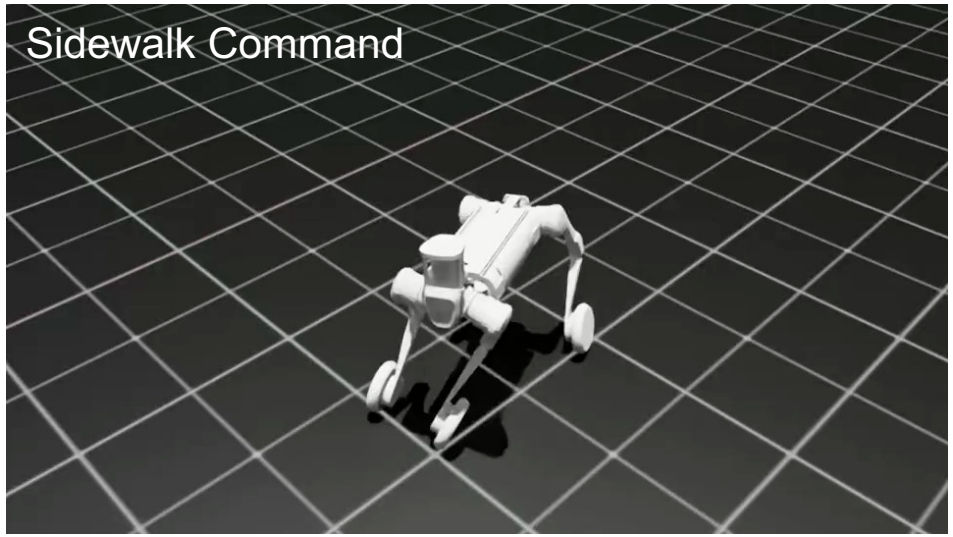
Trained Behaviors

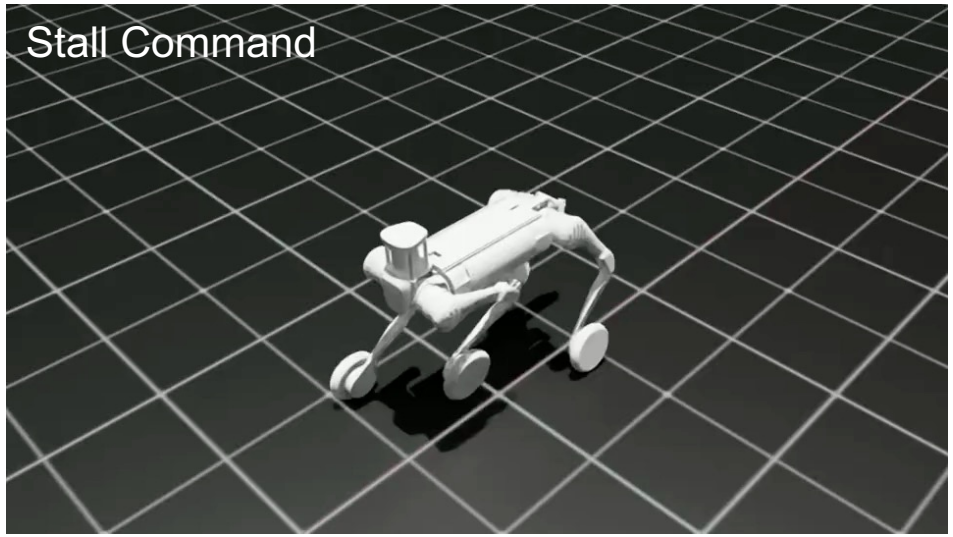
Robot shows locomotion with wheels

Results: Hybrid Locomotion



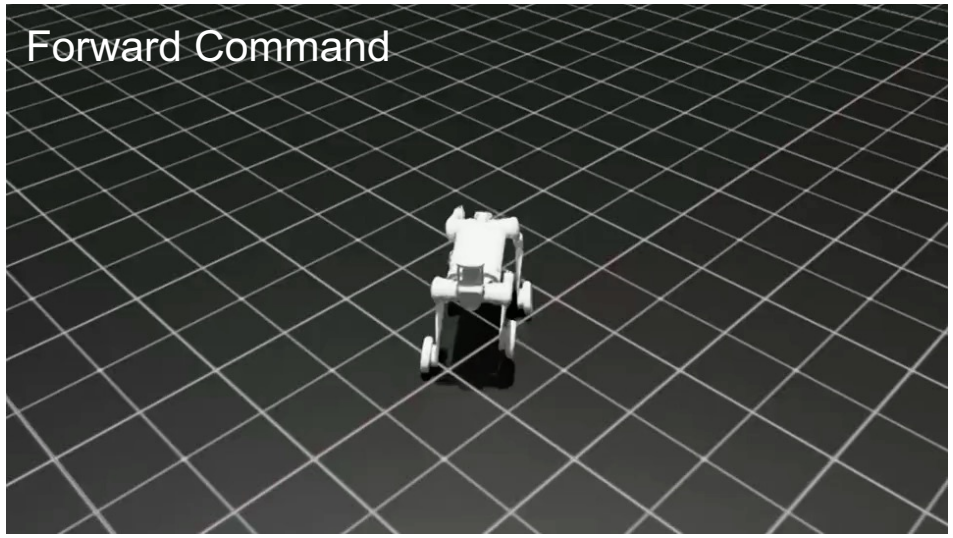


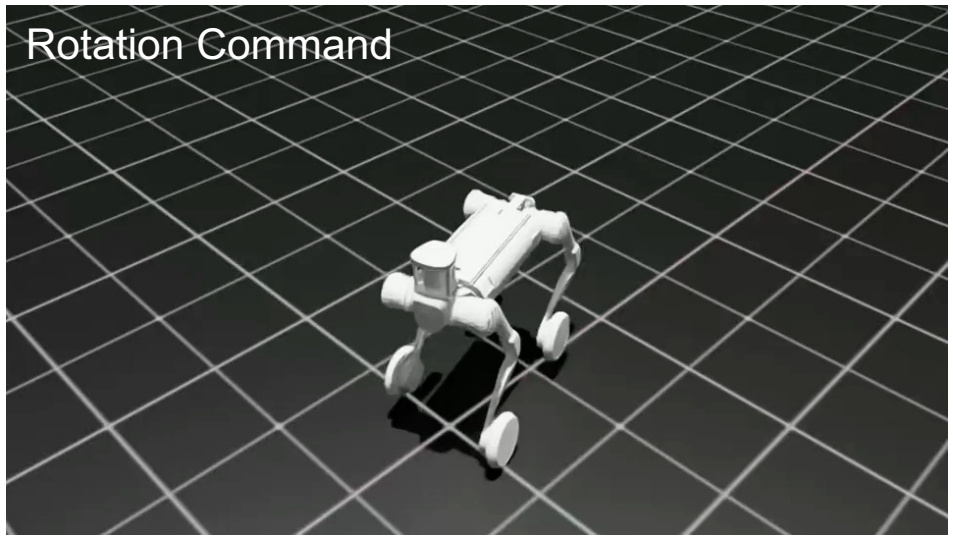


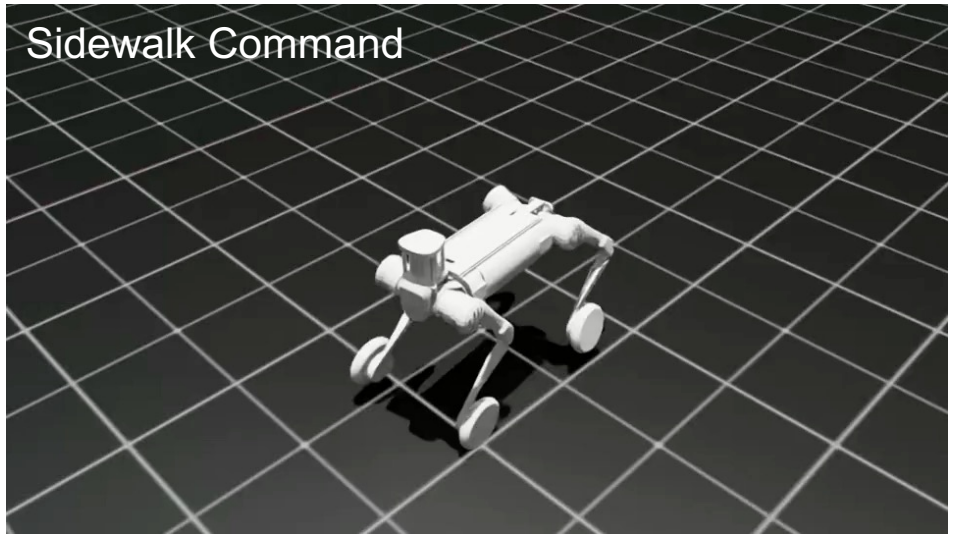


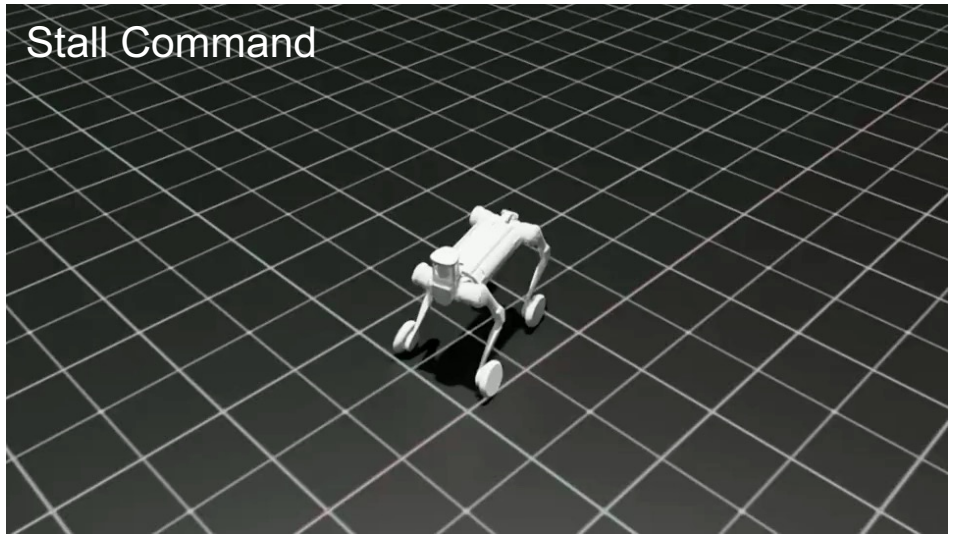
Results: Walking (Locked Wheels)

$$\omega_{i,target} = 0$$



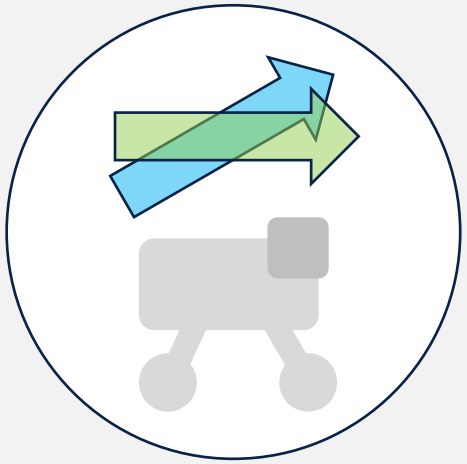






Inducing Different Motion

Command



Unlimtied Velocity Rewards

Miki et al. (2022)

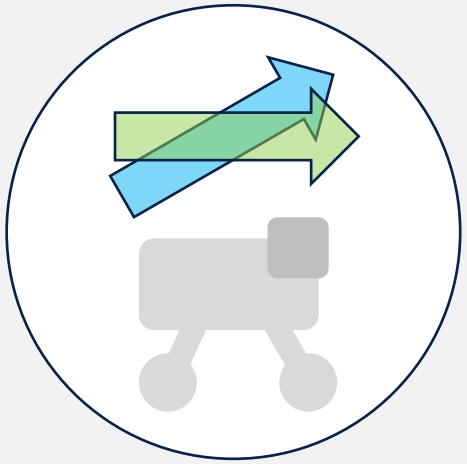
$$R_{vel} \in [0, 1]$$

New reward

$$R'_{vel} \propto proj(vel)$$
$$R'_{vel} \in [0, \infty)$$

Inducing Different Motion

Command



Unlimtied Velocity Rewards

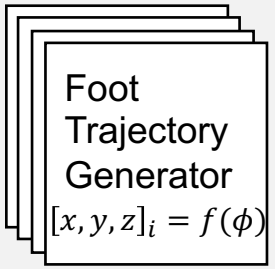
Miki et al. (2022)

$$R_{vel} \in [0, 1]$$

New reward

$$R'_{vel} \propto proj(vel)$$
$$R'_{vel} \in [0, \infty)$$

FTG

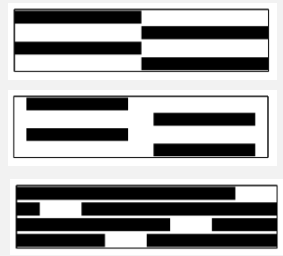


Phase Update

$$\phi_t(swing) = \phi_{t-1} + f\Delta t$$
$$\phi_t(stall) = \phi_{t-1} + (\lambda \cdot f)\Delta t$$

- Walking $(\lambda = 1)$
- Running $(\lambda > 1)$
- Strolling $(\lambda < 1)$

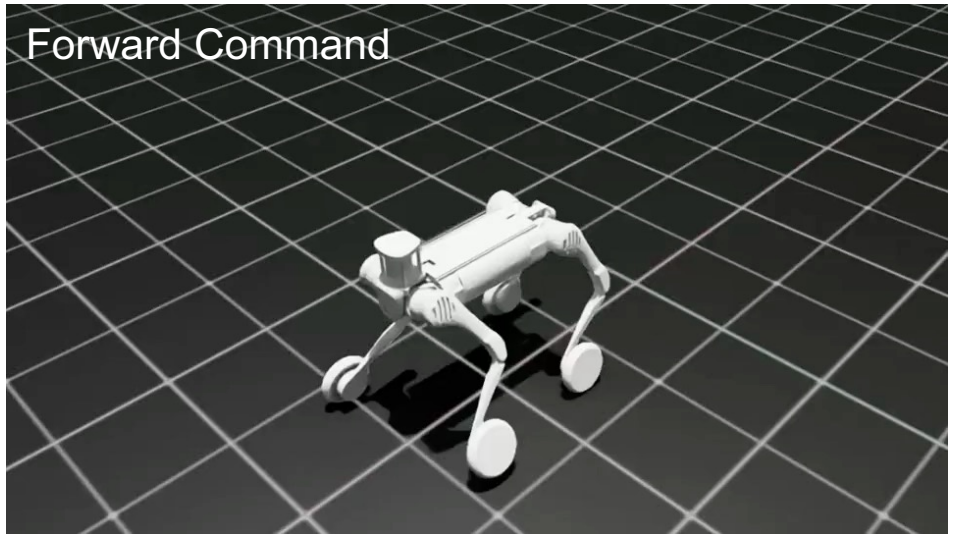
1)

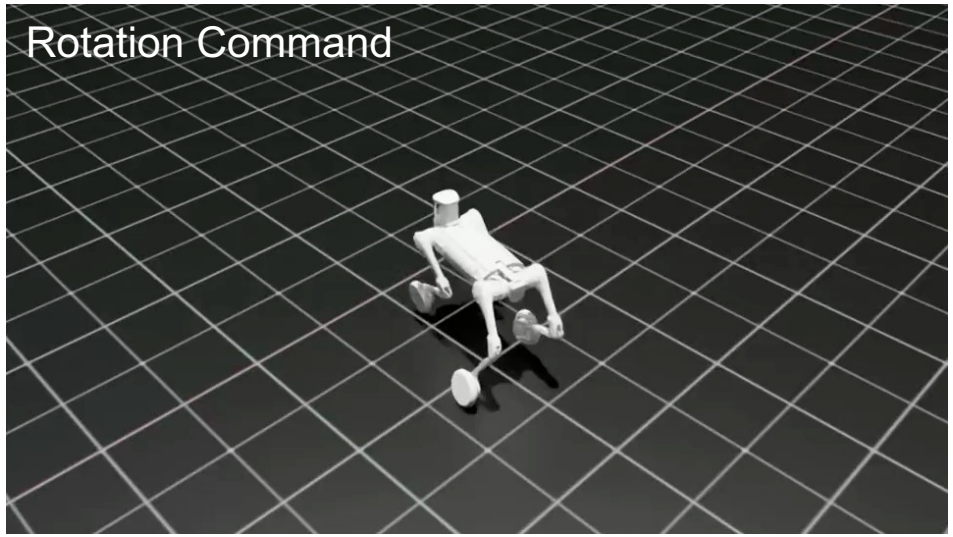


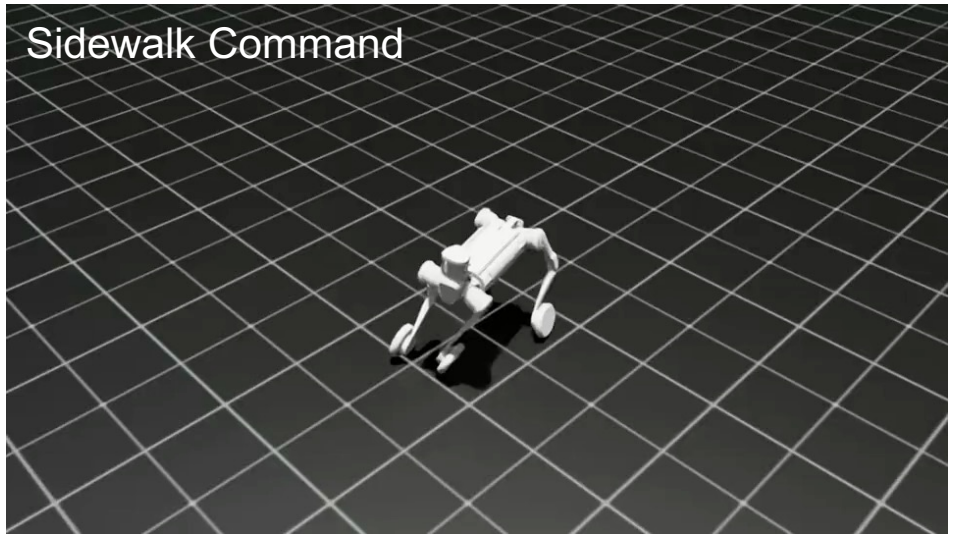
1. Control of Dynamic Gaits for a Quadrupedal Robot, Christian Gehring et al., (2013)

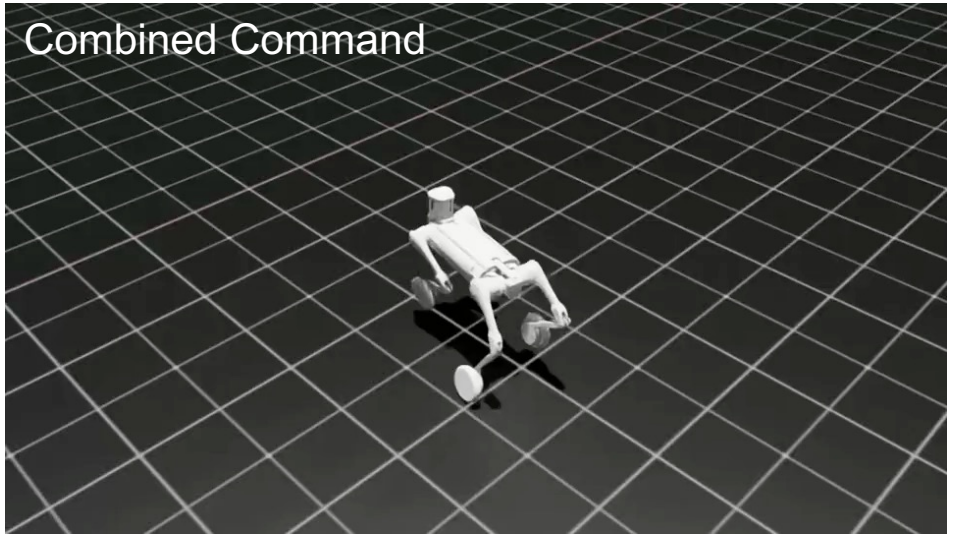
Results: Hybrid Locomotion (Walking)
Unlimited Speed Reward

Command reward proportional to velocity



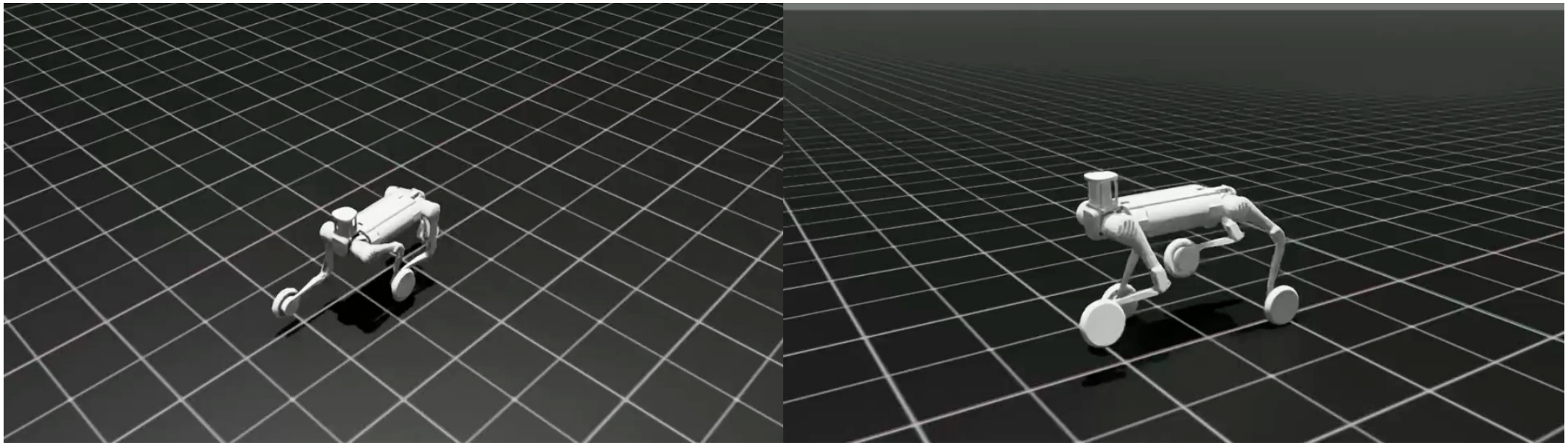






Results: Hybrid Locomotion (Running)
Unlimited Speed Reward

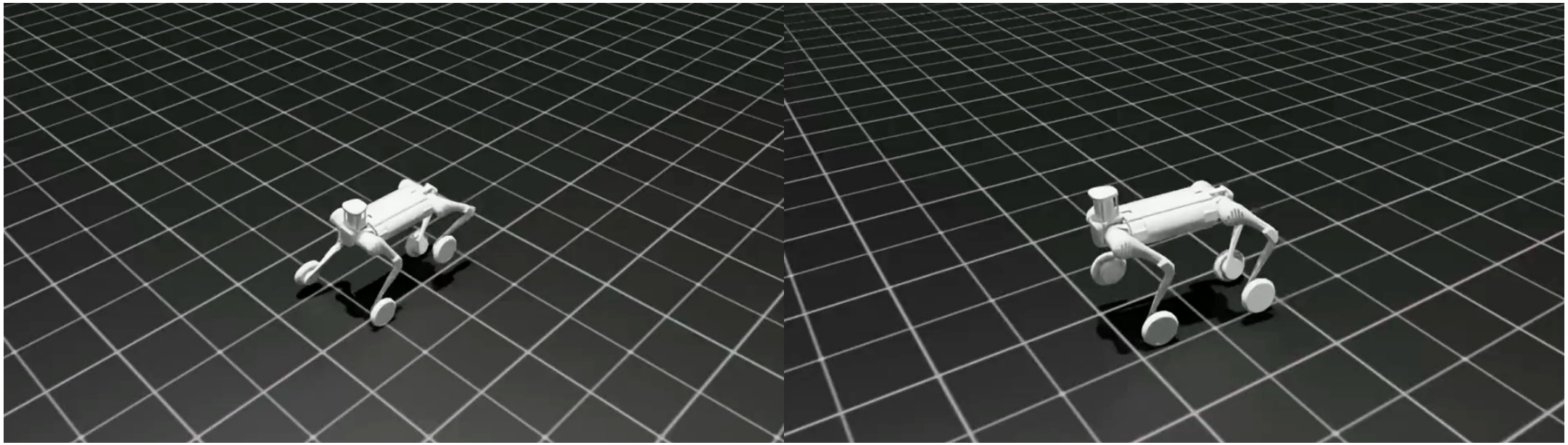
Running



- Command reward proportional to velocity
- Gait frequency multiplier = 2 (faster kick)

Results: Hybrid Locomotion (Strolling)
Unlimited Speed Reward

Strolling



- Command reward proportional to velocity
- Gait frequency multiplier = 0.5 (slower kick)

Conclusion

In this project,

- We explored the state-of-the-art method to control wheel legged locomotion.
- We implemented a learning pipeline for Unitree B2W robot.
- We discovered diverse wheeled legged locomotion behaviors.

Future work

The further steps are …

- Hardward experiment
  - Implement student policy training system (student - teacher approach)
- Terrain adaptive behaviors for enhanced mobility
  - Hill, stairs, bumpy floors …
- Incorporating high level navigation policy

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